

Microwave Impedance Microscopy: Circuit Physics and Simulation

The reflectometry signal chain, from Maxwell-scale tip-sample admittance to the demodulated MIM-Re / MIM-Im channels — derived, and reproduced with the `mimsim` scattering-network simulator.

Abstract. Microwave impedance microscopy (MIM) measures the complex admittance between a sharp tip and a sample at GHz frequencies, from which the local conductivity and permittivity are inferred. This tutorial builds the full measurement chain from first principles: the narrowband (complex-envelope) description of the signals, the scattering-wave formulation that turns the whole microwave circuit into one linear solve, the physics and S-matrix of every component, the impedance match that makes the tiny tip-sample perturbation observable, the three-arm architecture with carrier cancellation, and the IQ demodulation that produces the two MIM channels. Every result is reproduced numerically with `mimsim`.

1 · What MIM measures

A biased tip held just above a sample exchanges a small displacement current with it. The material responds through a *generalized complex permittivity*

$$\hat{\epsilon}(\mathbf{r}) = \epsilon + \frac{i\sigma}{\omega},$$

where ϵ is the permittivity, σ the (ac) conductivity — including free-carrier conduction and dielectric loss — and $\omega = 2\pi f$ the angular frequency. The quantity the instrument actually senses is the *tip-sample admittance* $\mathbf{Y}_{ts}$ (the inverse of impedance): its imaginary part stores reactive energy (capacitance) and its real part dissipates power. The central experimental fact is that $\mathbf{Y}_{ts}$ is tiny — of order attofarads (10^{-18} F) — compared with the probe's own self-capacitance of ~ 1 pF. MIM is, at its heart, the art of measuring that small change on a large background.

Because $\mathbf{Y}_{ts}$ is measured through the *reflection* of a microwave sent down a transmission line to the tip, the natural language is that of scattering waves. We develop it next.

2 · Narrowband signals: the complex envelope

Every signal in the instrument is a carrier at f_0 (typically 0.1–10 GHz) modulated slowly — by tip-height oscillation, drift, or a gate sweep — at kHz rates. We write a real voltage as

$$v(t) = \text{Re}\{a(t)e^{i\omega_0 t}\},$$

with $a(t)$ the slowly-varying *complex envelope*. Amplitude and phase live in $|a|$ and $\arg a$. This separates the two timescales: the GHz carrier need never be sampled directly (which over a millisecond would demand $\sim 10^{12}$ points), while the envelope is sampled only fast enough to follow the modulation. A pure tone has constant a ; a height-modulated tip gives an $a(t)$ whose spectrum is a carrier line plus sidebands at $f_0 \pm f_{\text{mod}}$ (Section 9). In `mimsim` this is the `Signal` object, with `.waveform()` reconstructing $v(t)$ and `.spectrum()` giving the narrowband RF spectrum.

3 · Scattering waves and the network solve

On a transmission line of reference impedance Z_0 (usually 50 Ω), it is convenient to split the voltage and current at any port into a forward (incident) wave a and a backward (reflected) wave b :

$$a = \frac{V + Z_0 I}{2\sqrt{Z_0}}, \quad b = \frac{V - Z_0 I}{2\sqrt{Z_0}} \iff V = \sqrt{Z_0}(a + b), \quad I = \frac{a - b}{\sqrt{Z_0}}.$$

The time-averaged power flowing into the port is $P = \frac{1}{2}(|a|^2 - |b|^2)$. A one-port load is fully described by its *reflection coefficient* $\Gamma = b/a$, related to its impedance Z or admittance $Y = 1/Z$ by

$$\Gamma = \frac{Z - Z_0}{Z + Z_0} = \frac{Y_0 - Y}{Y_0 + Y}, \quad Y_0 = \frac{1}{Z_0}.$$

An N -port component maps incident to outgoing waves through its *scattering matrix* $\mathbf{S}$, plus a source term for active parts:

$$\mathbf{b} = \mathbf{S} \mathbf{a} + \mathbf{b}_{\text{src}}.$$

An ideal connection between two ports simply exchanges their waves — what leaves one enters the other, $\mathbf{a}_i = \mathbf{b}_j$. Collecting *all* ports of the whole circuit into global vectors and writing the connection map as a permutation matrix $\mathbf{P}$ (so that $\mathbf{a} = \mathbf{P} \mathbf{b}$), the entire microwave network becomes a single linear system:

$$\mathbf{b} = \mathbf{S}(\mathbf{P} \mathbf{b}) + \mathbf{b}_{\text{src}} \implies \boxed{(\mathbf{I} - \mathbf{S} \mathbf{P}) \mathbf{b} = \mathbf{b}_{\text{src}}}.$$

Solving this at the carrier frequency yields every wave at every port at once — so reflections, impedance mismatch, standing waves and cancellation are all handled exactly, not approximated. Unconnected ports are treated as matched terminations ($\mathbf{a}_i = \mathbf{0}$). Slow modulation is handled by re-solving on a slow-time grid, because a control voltage or the tip height enters $\mathbf{S}$ through the affected blocks. This boxed equation is the engine of the simulator ([Network.run](#)).

4 · The building blocks and their S-matrices

4.1 Source

A matched source is a one-port that emits a wave and reflects returning power through its internal reflection Γ_s (ideally $\mathbf{0}$): $\mathbf{b} = \Gamma_s \mathbf{a} + \mathbf{b}_s$, with $\mathbf{b}_s = A e^{i\varphi}$. Its available power is $P = \frac{1}{2} |b_s|^2$, so a level in dBm sets $|b_s| = \sqrt{2 \cdot 10^{-3} 10^{P_{\text{dBm}}/10}}$. The source amplitude, phase and frequency f_0 are the tunable inputs.

4.2 Directional coupler

A four-port coupler (ports *in*, *through*, *coupled*, *isolated*) samples a defined fraction of the forward wave while ideally isolating the reverse wave — the key to separating the reflected signal from the incident one. With voltage coupling $k = 10^{-C_{\text{dB}}/20}$ and through transmission $t = \sqrt{1 - k^2}$, the ideal scattering matrix is

$$\mathbf{S} = \begin{pmatrix} 0 & t & ik & 0 \\ t & 0 & 0 & ik \\ ik & 0 & 0 & t \\ 0 & ik & t & 0 \end{pmatrix}.$$

The coupled port carries the $+90^\circ$ factor i . Finite directivity is modelled by a small leakage into the isolated port. A wave entering *through* (a reflection returning from the tip) emerges mostly at the *isolated* port — which is exactly where the amplifier taps it.

4.3 Voltage-controlled attenuator and phase shifter

These two-ports are matched and reciprocal; only their transmission carries information. An applied control voltage V maps through a user-defined response to an attenuation $A(V)$ in dB or a phase $\phi(V)$ in radians:

$$\mathbf{S}_{\text{att}} = \begin{pmatrix} 0 & \tau \\ \tau & 0 \end{pmatrix}, \quad \tau = 10^{-A(V)/20}; \quad \mathbf{S}_{\text{ps}} = \begin{pmatrix} 0 & e^{i\phi(V)} \\ e^{i\phi(V)} & 0 \end{pmatrix}.$$

These are the knobs of the cancellation and reference arms (Section 8).

4.4 Amplifier

A unilateral amplifier is matched with reverse isolation, so only S_{21} is nonzero: $\mathbf{S} = \begin{pmatrix} 0 & 0 \\ G & 0 \end{pmatrix}$, with $G = 10^{G_{\text{dB}}/20} e^{i\varphi}$. It raises the (cancelled) reflected signal before demodulation.

4.5 Two-port matching network (ABCD cascade)

A general lossless/lossy two-port is most easily built from cascaded transmission (ABCD) matrices — a series impedance is $\begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}$, a shunt admittance $\begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}$, a line of length ℓ is $\begin{pmatrix} \cos \beta \ell & i Z_\ell \sin \beta \ell \\ i \sin \beta \ell / Z_\ell & \cos \beta \ell \end{pmatrix}$. The cascade product $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ converts to scattering parameters via

$$S_{11} = \frac{A + B/Z_0 - CZ_0 - D}{\Delta}, \quad S_{21} = \frac{2}{\Delta}, \quad \Delta = A + B/Z_0 + CZ_0 + D.$$

Its role in MIM is developed in Section 7.

5 · Tip-sample admittance and the response curve

The tip couples to the sample through a small interface capacitance C_c ; below it, the material presents a resistance R in parallel with a capacitance C_s . Writing $R = \rho/g$ and $C_s = \epsilon g$ for a geometric length scale g , the lumped tip-sample admittance is

$$Y_{ts} = \left[\frac{1}{i\omega C_c} + \left(R \parallel \frac{1}{i\omega C_s} \right) \right]^{-1}, \quad R \parallel \frac{1}{i\omega C_s} = \frac{R}{1 + i\omega R C_s}.$$

The probe adds its self-capacitance in parallel, $Y = i\omega C_{\text{probe}} + Y_{ts}$, so $Y_{ts} \ll Y$ is a perturbation. Carrying out the algebra, the dissipative (real) part is

$$\text{Re } Y_{ts} = \frac{\omega^2 C_c^2 G_s}{G_s^2 + \omega^2 (C_s + C_c)^2}, \quad G_s = 1/R,$$

which, maximized over G_s , peaks at $G_s = \omega(C_s + C_c)$. Since $RC_s = \rho\epsilon$ is geometry-independent, the dissipation peak sits at

$$\rho\omega\epsilon = \frac{1}{1 + C_c/C_s} \xrightarrow{C_c \ll C_s} 1,$$

i.e. at $\sigma \sim \omega\epsilon$: the sample's charge-relaxation rate matches the measurement frequency. Physically, at low σ the field penetrates and little is lost; at high σ the field is screened and the sample acts as a ground plane (large Im , small Re); in between, the $\mathbf{E} \cdot \mathbf{J}$ overlap is largest and dissipation peaks. This is the celebrated two-decade contrast window (Fig. 1).

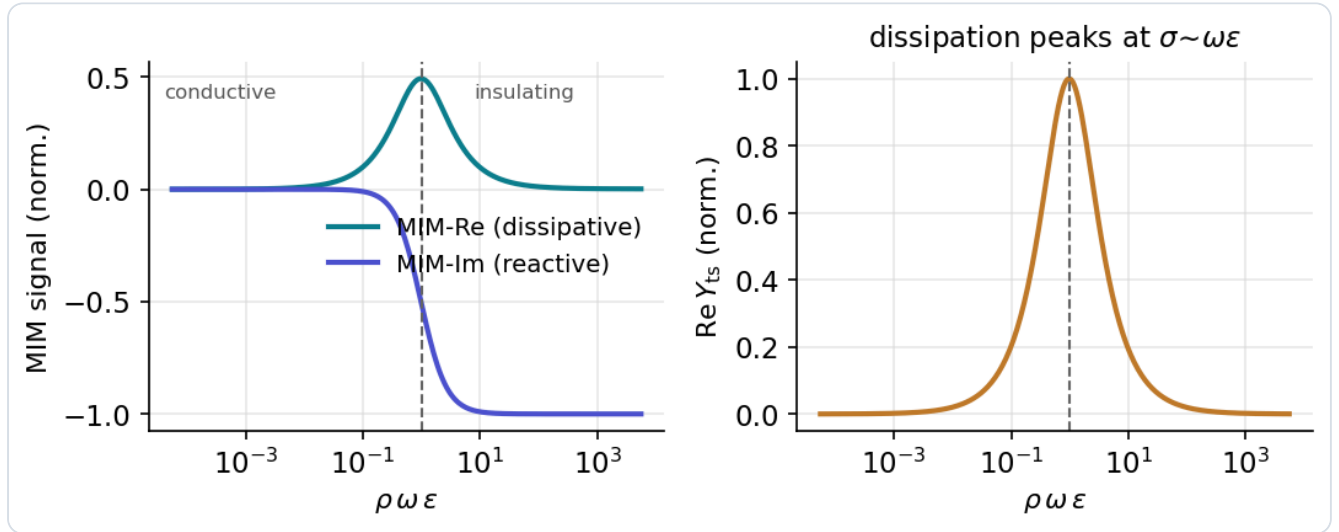


Figure 1. Simulated MIM response curve. Left: the demodulated MIM-Re (dissipative) and MIM-Im (reactive) channels versus the dimensionless $\rho\omega\epsilon$, after LO-phase calibration (Section 6). MIM-Re peaks near $\rho\omega\epsilon \approx 1$ and MIM-Im rises monotonically toward the conductive side. Right: the modelled conductance $\text{Re } Y_{ts}$, confirming the dissipation peak at $\sigma \sim \omega\epsilon$.

6 · IQ demodulation: the two MIM channels

Because the tip-sample term is a small perturbation, the change in reflection is linear in it,

$$\delta\Gamma = \frac{d\Gamma}{dY} \delta Y_{ts},$$

with $d\Gamma/dY$ a fixed complex gain set by the probe and matching network. The reflected wave, amplified, arrives at the RF port of an IQ mixer, where it is multiplied against a phase-coherent reference (the LO) tapped from the same source. In envelope form the demodulated output is

$$\text{MIM-Re} + i \text{MIM-Im} = \frac{1}{2} a_{\text{RF}} \frac{\overline{a_{\text{LO}}}}{|a_{\text{LO}}|} \propto e^{-i\theta_{\text{LO}}} \delta\Gamma.$$

The LO phase θ_{LO} — set by the reference-arm phase shifter — is a genuine degree of freedom: it rotates the complex signal in the I/Q plane. One calibrates it so the *dissipative* response lands in MIM-Re and the *reactive* response in MIM-Im; otherwise the two channels mix. This calibration is exactly the rotation applied in Fig. 1.

7 · Impedance matching and sensitivity

A crucial subtlety: a *lossless* reactive probe (a pure ~ 1 pF capacitor) reflects *everything*, $|\Gamma| = 1$, no matter what lossless network precedes it — matching can only rotate the phase of Γ . A real match relies on the probe's finite quality factor Q . The paper's scheme, a $\lambda/4$ resonant match, works in two steps: a shunt inductor $L = 1/(\omega_0^2 C_{\text{probe}})$ resonates out the probe capacitance, turning the probe into a real resistance $R_{\text{loss}} = Q/(\omega_0 C_{\text{probe}})$; a quarter-wave line of impedance $Z_{\lambda/4} = \sqrt{Z_0 R_{\text{loss}}}$ then transforms R_{loss} down to Z_0 . At the match, $|\Gamma|$ dips and the sensitivity $|d\Gamma/dY|$ — hence the MIM signal — peaks (Fig. 2). What the network buys is dynamic range: for the near-lossless MIM probe, inserting the match increases the demodulated tip-sample contrast by about two orders of magnitude in the simulator. A perfect match is not required — a modest residual mismatch barely costs signal — but the resonator drifting off f_0 does reduce it, which is why drift stabilization matters.

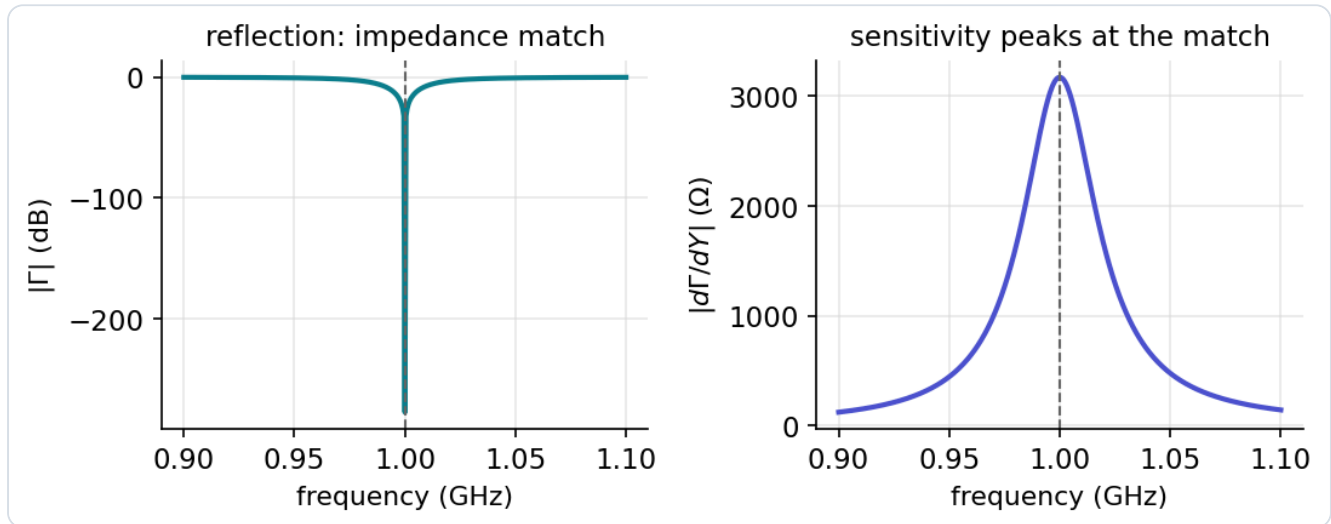


Figure 2. Resonant match. Left: $|\Gamma|$ versus carrier frequency for a 1 pF probe of quality factor $Q = 40$; the reflection dips at the resonant match. Right: the sensitivity $|d\Gamma/dY|$, which peaks precisely at the match — where a given tip-sample admittance change produces the largest change in reflection.

8 · The three-arm system and carrier cancellation

One source feeds three arms (Fig. 3). The **excitation** arm drives the tip through the matching network. The reflection returns through the excitation coupler's isolated port. The **cancellation** arm — a voltage-controlled attenuator A and phase shifter ϕ — synthesizes a second wave that meets the reflected wave at a combiner and nulls the carrier *before* the amplifier. The **reference** arm — a phase shifter — supplies the LO. If the combiner passes the reflected wave with transmission t and injects the cancellation wave with coupling ik , the amplifier input is $b_{\text{amp}} = t b_{\text{refl}} + ik b_{\text{cancel}}$, which vanishes when

$$b_{\text{cancel}} = \frac{it}{k} b_{\text{refl}},$$

i.e. for the cancellation amplitude and phase that make the injected wave equal and opposite to the reflected carrier. Nulling the large static carrier lets the amplifier run at high gain without saturating, so the tiny modulated tip-sample term survives to the mixer. In the simulator, tuning A and ϕ drives the carrier at the amplifier input into a deep null (Fig. 3, right).

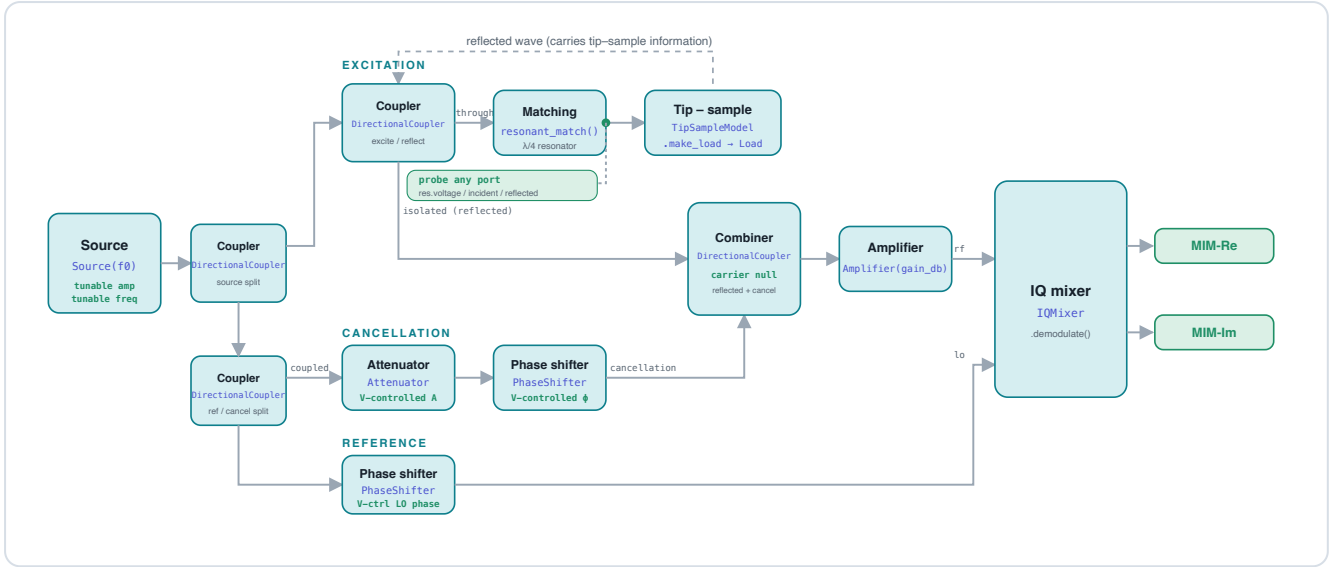


Figure 3a. The three-arm reflectometer (excitation / cancellation / reference), with each physical device labelled by its `mimsim` class and ports. The reflected wave meets the cancellation wave at the combiner.

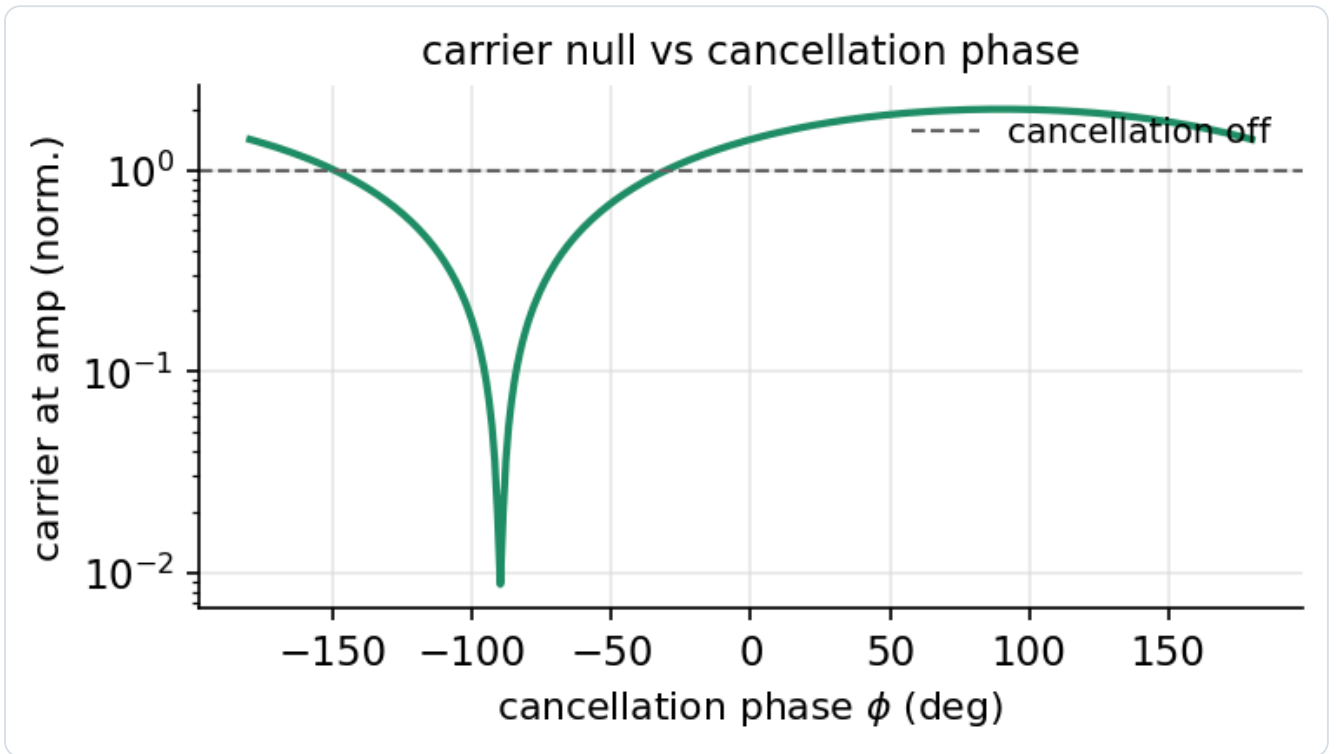


Figure 3b. Carrier null. Carrier amplitude at the amplifier input, normalized to the un-cancelled case, as the cancellation phase ϕ is swept at the tuned attenuation. At the correct phase the reflected carrier is deeply nulled.

9 · Modulation and lock-in detection

Rather than measure the absolute reflection, MIM typically *modulates* — oscillating the tip height $z(t)$ at f_{mod} , which modulates C_c and hence Y_{ts} . In the RF spectrum this places sidebands at $f_0 \pm f_{\text{mod}}$; after the mixer, the modulation appears as a clean baseband line at f_{mod} (Fig. 4). A lock-in at f_{mod} then extracts the $d\text{MIM}/dz$ signal while rejecting slow drift — the same reason the raw carrier, which is enormous compared with the tip-sample perturbation, must first be cancelled and then demodulated.

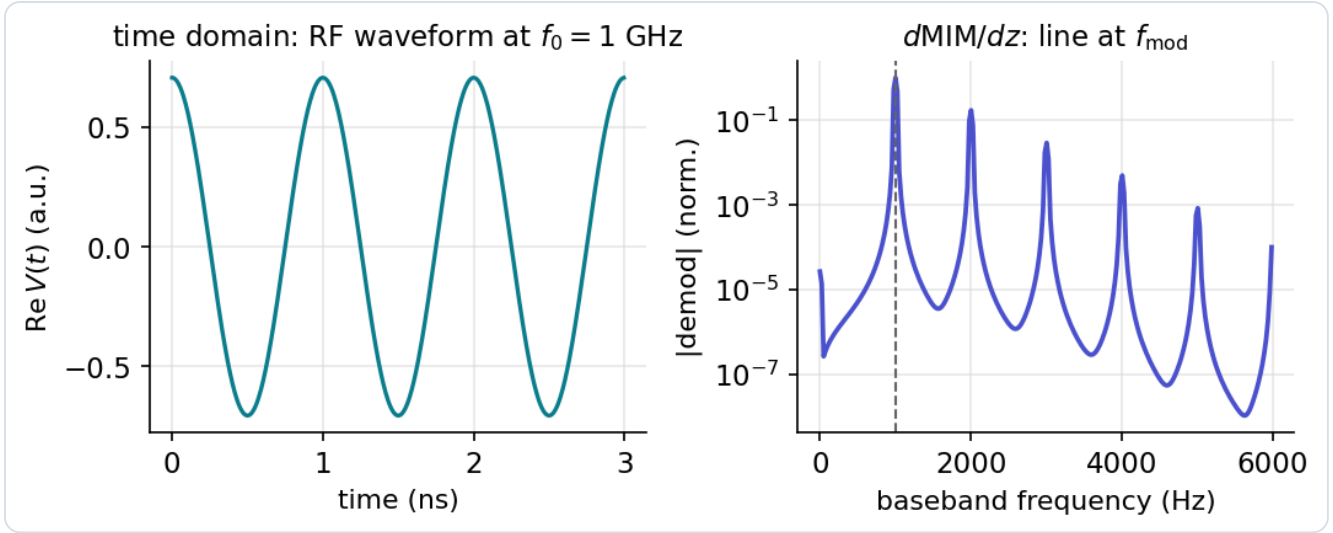


Figure 4. Signal in time and frequency. Left: the reconstructed RF voltage $\text{Re } V(t)$ at a node, oscillating at $f_0 = 1$ GHz. Right: with the tip height modulated at $f_{\text{mod}} = 1$ kHz, the demodulated baseband spectrum shows a clean line at f_{mod} — the lock-in observable, well above the numerical floor.

10 · Physics ↔ equation ↔ code

PHYSICAL STEP	GOVERNING RELATION	MIMSIM
Solve the whole circuit	$(\mathbf{I} - \mathbf{SP})\mathbf{b} = \mathbf{b}_{\text{src}}$	<code>Network.run</code>
One-port reflection	$\Gamma = (Y_0 - Y)/(Y_0 + Y)$	<code>Load</code> , <code>Termination</code>
Voltage-controlled A / φ	$\tau = 10^{-A(V)/20}$, $e^{i\phi(V)}$	<code>Attenuator</code> , <code>PhaseShifter</code>
Resonant match	$Z_{\lambda/4} = \sqrt{Z_0 R_{\text{loss}}}$	<code>resonant_match</code>
Tip-sample admittance	$\text{Re } Y_{\text{ts}}$ peaks at $\sigma \sim \omega \epsilon$	<code>TipSampleModel</code>
IQ demodulation	$\propto e^{-i\theta_{\text{LO}}} \delta \Gamma$	<code>IQMixer.demodulate</code>
Carrier cancellation	$b_{\text{cancel}} = \frac{i\epsilon}{k} b_{\text{refl}}$	cancellation arm
Signal in time / frequency	$v = \text{Re}\{a e^{i\omega_0 t}\}$	<code>Signal.waveform</code> / <code>.spectrum</code>

Appendix · Reference parameters

The figures use $f_0 = 1$ GHz, sample permittivity $\epsilon = 10 \epsilon_0$, probe self-capacitance $C_{\text{probe}} = 1$ pF and quality factor $Q = 40$ (so $L = 25.3$ nH, $R_{\text{loss}} = 6.4$ k Ω , $Z_{\lambda/4} = 564 \Omega$, a 7.5 cm quarter-wave line at 1 GHz), an interface capacitance C_e small compared with the sample capacitance $C_s = \epsilon g$ (placing the dissipation peak at $\rho \omega \epsilon \approx 1$), a 10 dB excitation coupler and a 30 dB amplifier. The height modulation is $f_{\text{mod}} = 1$ kHz with a 0–10 nm gap. All values are set in `docs/tutorial_figures.py`.